

Sri Lanka Institute of Information Technology



Kandy Uni

**IT2011 - Artificial Intelligence and Machine
Learning**

Lab Submission 06

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Task 1– K-Means Clustering

```
[ ] #Task 1
    # Step 1: Import libraries

    import pandas as pd
    import matplotlib.pyplot as plt
    from sklearn.cluster import KMeans
    from sklearn.datasets import load_iris
```

```
[ ] # Step 2: Load dataset

    iris = load_iris()

    df = pd.DataFrame(iris.data, columns=iris.feature_names)

    print(df.head())
```

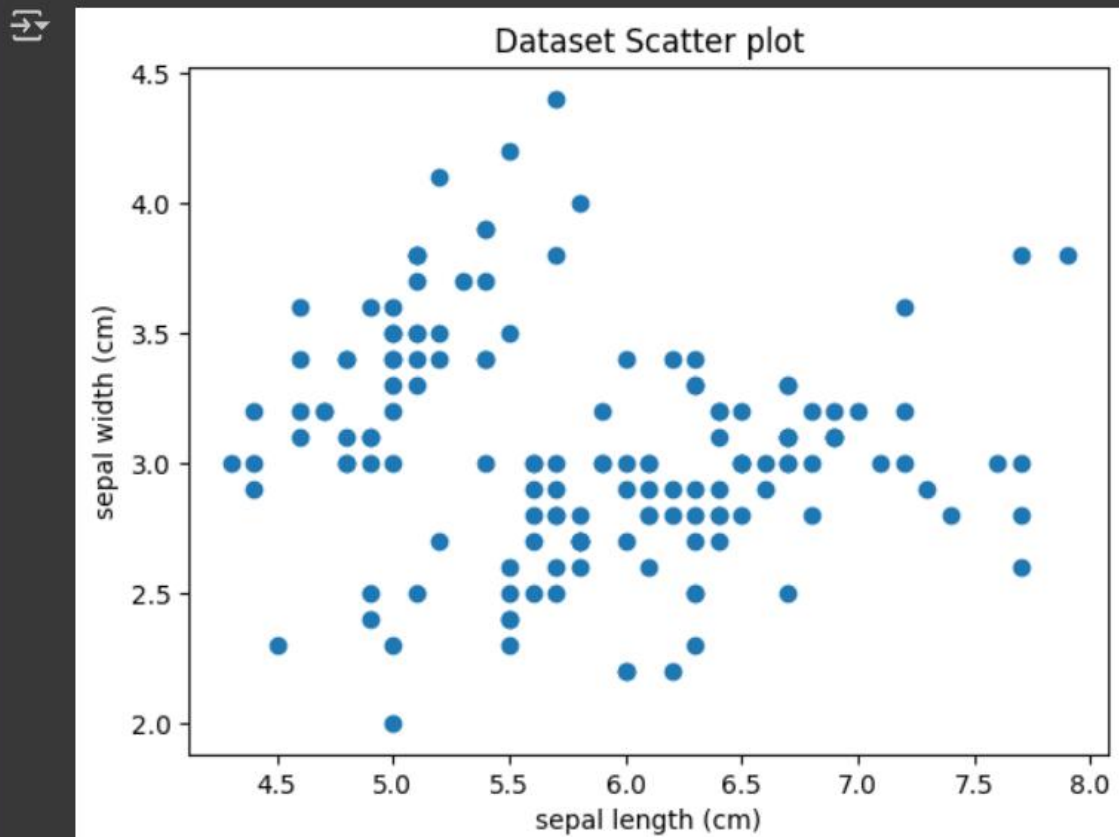
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

```
# Step 3:relevant features

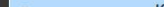
features=df[['sepal length (cm)','sepal width (cm)']]
```

4. Visualize the dataset using a scatter plot

```
plt.scatter(features['sepal length (cm)'], features['sepal width (cm)'])
plt.xlabel('sepal length (cm)')
plt.ylabel('sepal width (cm)')
plt.title('Dataset Scatter plot')
plt.show()
```



```
kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(features)
```



```
KMeans(n_clusters=3, random_state=42)
```

▶ # Step 6: Retrieve the cluster labels

```
labels = kmeans.labels_  
print(labels)
```

$$\Sigma = \begin{bmatrix} 2 & 2 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Step 7: Add these cluster labels as a new column to the dataset

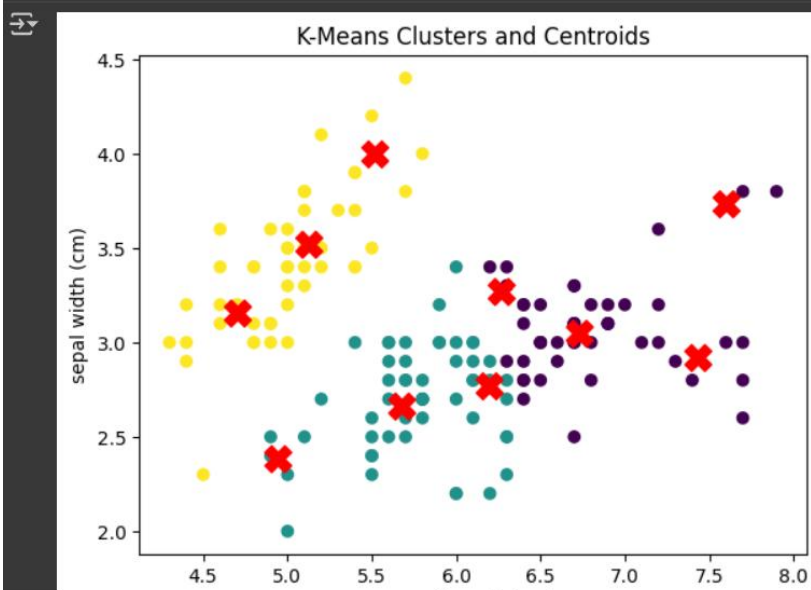
```
df['Cluster'] = labels
print(df.head())
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

	Cluster
0	2
1	2
2	2
3	2
4	2

8. Visualize the clusters and their centroids using a scatter plot

```
plt.scatter(features['sepal length (cm)'], features['sepal width (cm)'], c=labels, cmap='viridis')
centroids=kmeans.cluster_centers_
plt.scatter(centroids[:, 0], centroids[:, 1], marker='x', s=200, c='red', label='centroids' )
plt.xlabel('sepal length (cm)')
plt.ylabel('sepal width (cm)')
plt.title('K-Means Clusters and Centroids')
plt.show()
```



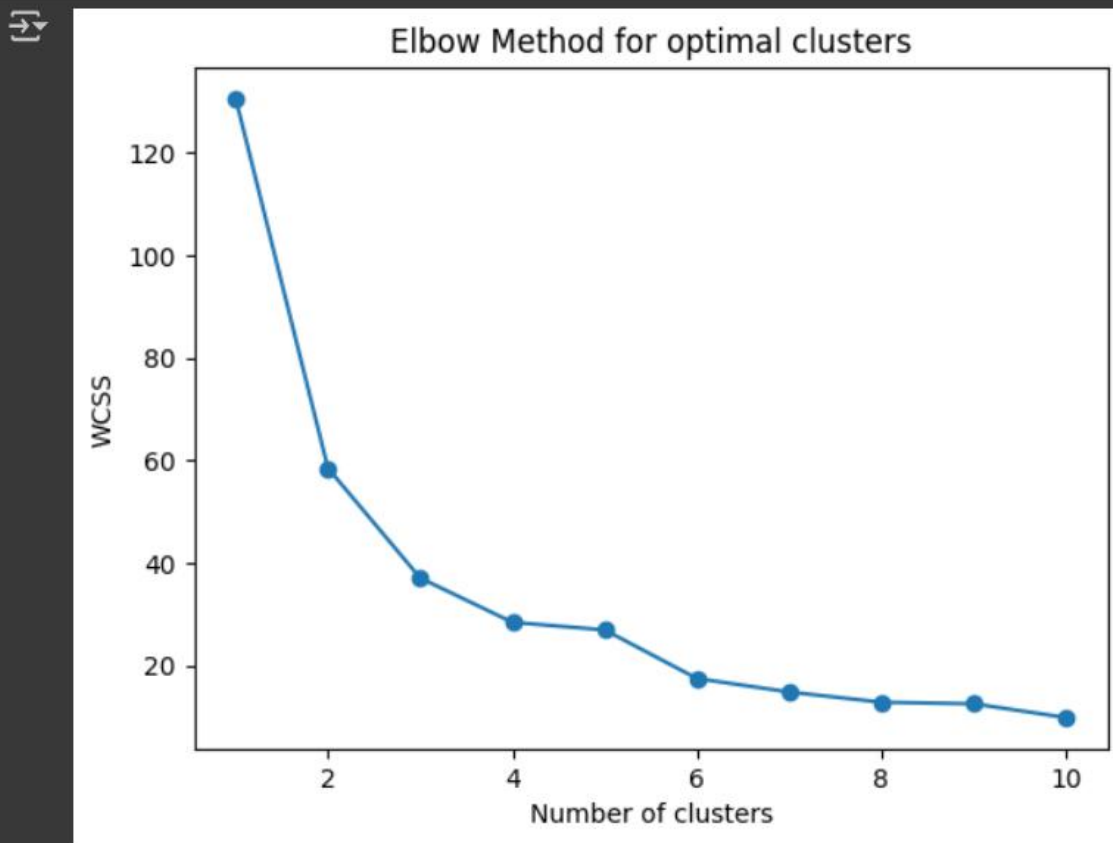
Step 9: Calculate the Within-Cluster Sum of Squares

```
wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(features)
    wcss.append(kmeans.inertia_)
print(wcss)
```

```
[138.475266666667, 58.4475924688194, 37.0507021276596, 28.41365317351379, 26.9325075582776, 17.463509575220108, 14.828543464836567, 12.843135057471263, 12.530893304970899, 9.854
```

10. Use the Elbow Method to determine the optimal number of clusters

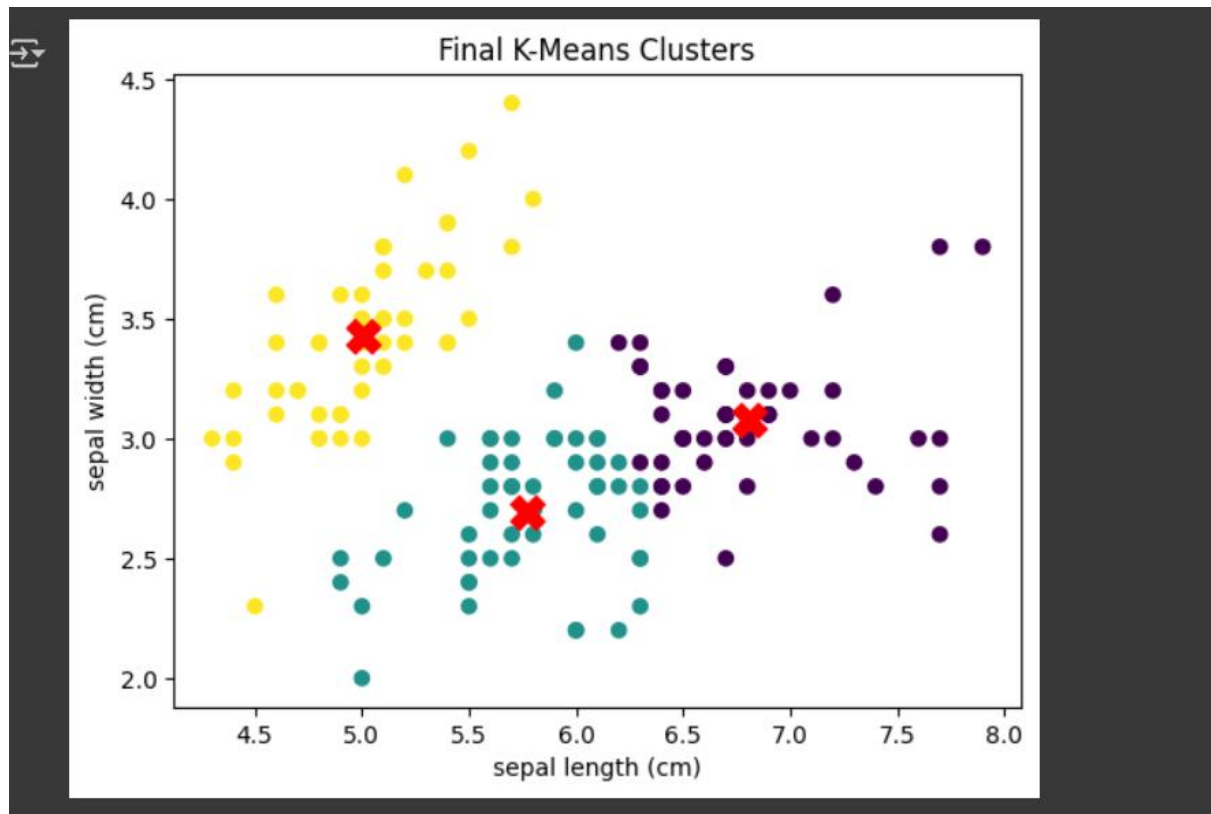
```
plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.title('Elbow Method for optimal clusters')
plt.show()
```



```
#Step 11:Re-train the K-Means model with the optimal number of clusters and plot the final clusters

optimal_k=3
kmeans_optimal=KMeans(n_clusters=optimal_k,random_state=42)
kmeans_optimal.fit(features)
labels_optimal=kmeans_optimal.labels_
df['optimal_cluster']=labels_optimal

plt.scatter(features['sepal length (cm)'],features['sepal width (cm)'],c=labels_optimal,cmap='viridis')
centroids_optimal=kmeans_optimal.cluster_centers_
plt.scatter(centroids_optimal[:,0],centroids_optimal[:,1],marker='x',s=200,c='red',label='centroids')
plt.xlabel('sepal length (cm)')
plt.ylabel('sepal width (cm)')
plt.title('Final K-Means Clusters')
plt.show()
```



Task 2– Hierarchical Clustering

```
# Task 2 :
# Step 12: Import libraries for hierarchical clustering and visualization

from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage
```

```
# Step 13: Load dataset again
```

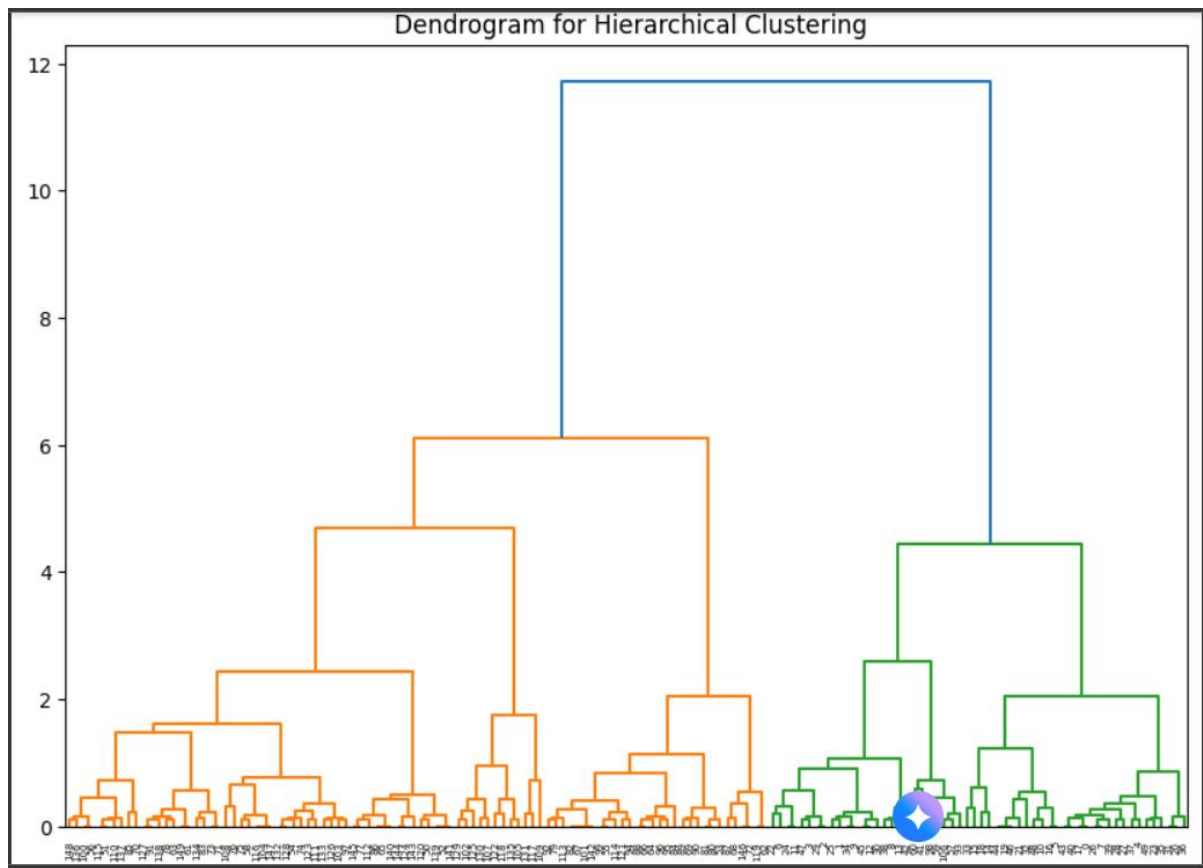
```
print("Dataset preview:")
print(df.head())
```

```
Dataset preview:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1                3.5                1.4                0.2
1                4.9                3.0                1.4                0.2
2                4.7                3.2                1.3                0.2
3                4.6                3.1                1.5                0.2
4                5.0                3.6                1.4                0.2

   Cluster  optimal_cluster
0         2                2
1         2                2
2         2                2
3         2                2
4         2                2
```

```
features = df[['sepal length (cm)', 'sepal width (cm)']]
```

```
linked = linkage(features, method='ward')
plt.figure(figsize=(10,7))
dendrogram(linked, orientation='top', distance_sort='descending', show_leaf_counts=True)
plt.title('Dendrogram for Hierarchical Clustering')
plt.show()
```



```
hc = AgglomerativeClustering(n_clusters=3, linkage='ward')
hc.fit(features)
```

```
hc_labels = hc.labels_  
print("Hierarchical Cluster Labels:", hc_labels)
```

[illegible]

```
# Step 18: Add the hierarchical cluster labels to the DataFrame
```

```
df['hierarchical_cluster'] = hc_labels  
print(df.head())
```

```
sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \  
0                5.1                3.5                1.4                0.2  
1                4.9                3.0                1.4                0.2  
2                4.7                3.2                1.3                0.2  
3                4.6                3.1                1.5                0.2  
4                5.0                3.6                1.4                0.2  
  
Cluster  optimal_cluster  hierarchical_cluster  
0        2                2                    1  
1        2                2                    1  
2        2                2                    1  
3        2                2                    1  
4        2                2                    1
```

```
# Step 19: Visualize final hierarchical clusters
```

```
plt.scatter(features['sepal length (cm)'], features['sepal width (cm)'], c=hc_labels, cmap='viridis')  
plt.xlabel('sepal length (cm)')  
plt.ylabel('sepal width (cm)')  
plt.title('Hierarchical Clustering Results')  
plt.show()
```

